# Post-Resuscitation Care

**AC-10 | Adult Cardiac**

## Quick flow

**01 Confirm Sustained ROSC [EMT / AEMT / Paramedic]**

Pulse • BP/MAP • organized rhythm • perfusion • keep pads attached • prepare for re-arrest

**02 Airway + Oxygenation [EMT / AEMT / Paramedic]**

100% oxygen until reliable $SpO_2$ • then titrate to 90-98% • waveform capnography for advanced airway

**03 Control Ventilation [EMT / AEMT / Paramedic]**

Begin near 10 breaths/min • target $EtCO_2$ approximately 35-45 mm Hg • avoid hyperventilation

**04 MAP ≥65 mm Hg? [AEMT / Paramedic]**

If low: NS 250-500 mL IV/IO and reassess • smaller volume for CHF, renal/liver failure, or overload

**05 Persistent Hypotension [Paramedic]**

Norepinephrine 0.1-2 mcg/kg/min IV/IO • titrate to MAP ≥65 • alternatives require Medical Control

**06 12-Lead ECG + Transmit [EMT / AEMT / Paramedic]**

Obtain as soon as feasible • repeat if immediate post-ROSC tracing or condition changes • open AC-04 when indicated

**07 Glucose + Neuro + Temperature [EMT / AEMT / Paramedic]**

Treat glucose <70 • document GCS/pupils/seizures • prevent fever • no rapid cold-fluid cooling

**08 Destination + Notify [Paramedic]**

PCI-capable center for STEMI, cardiogenic shock, recurrent ventricular arrhythmia, or ongoing ischemia • consider Covenant Health Air

## Overview

- Adult pathway for patients 16 years of age and older with sustained return of spontaneous circulation after cardiac arrest.
- Prioritize controlled oxygenation and ventilation, avoidance of hypotension, rapid 12-lead ECG acquisition, cause-specific treatment, appropriate destination, and preparation for re-arrest.
- Keep defibrillation pads attached throughout transport because recurrent arrest and ventricular dysrhythmias are common.

## EMT Care

Immediate post-ROSC stabilization, ventilation, oxygen titration, monitoring, neurologic assessment, and re-arrest preparation.

**EMT**

- Confirm sustained ROSC using a palpable pulse, blood pressure, organized rhythm, perfusion, and clinical response.
- Maintain airway and ventilation, begin near 10 breaths/min, and avoid hyperventilation.
- Use 100% oxygen until $SpO_2$ can be measured reliably, then titrate oxygen to maintain $SpO_2$ 90-98%.
- Keep defibrillation pads attached, obtain frequent complete vital signs, and prepare to resume CPR and defibrillation immediately.
- Check blood glucose, document GCS, pupils, purposeful movement and seizure activity, and measure temperature when available.
- Acquire and transmit a 12-lead ECG when trained and equipped without delaying stabilization or transport.

## AEMT Care

All EMT care plus vascular access, cautious fluid resuscitation, airway support, and medication preparation.

**AEMT**

- Perform all EMT actions and establish IV/IO access without delaying transport.
- For MAP below 65 mm Hg, give normal saline 250-500 mL IV/IO and reassess blood pressure, lung sounds, perfusion, and respiratory status.
- Use smaller fluid volumes for CHF, renal failure, liver failure, or suspected volume overload.
- Assist with advanced-airway confirmation and continuous waveform capnography; reassess breath sounds, chest rise, tube depth, and waveform after every movement.
- Prepare norepinephrine and recurrent-dysrhythmia medications for paramedic use.
- Recheck glucose and complete vital signs after interventions or any clinical change.

## Paramedic Care

All AEMT care plus advanced airway/capnography, vasopressor support, rhythm management, ECG interpretation, and destination selection.

**Paramedic**

- Perform all EMT and AEMT actions and identify the likely cause of arrest and immediate post-ROSC threats.
- Confirm an advanced airway with continuous waveform capnography and ventilate near 10 breaths/min, titrating toward $EtCO_2$ approximately 35-45 mm Hg while recognizing that $EtCO_2$ may underestimate $PaCO_2$ after ROSC.
- If MAP remains below 65 mm Hg after cautious fluid, start norepinephrine 0.1-2 mcg/kg/min IV/IO and titrate to MAP at least 65 mm Hg.
- Alternatives to norepinephrine for post-ROSC hypotension require Medical Control.
- Interpret and transmit the 12-lead ECG, repeat when indicated, and activate AC-04 for persistent STEMI/STEMI-equivalent or suspected ongoing ischemia.
- Treat recurrent ventricular dysrhythmia through the linked rhythm-specific protocol and prepare for immediate re-arrest.
- Choose the receiving destination based on STEMI, cardiogenic shock, recurrent ventricular arrhythmia, ongoing ischemia, transport time, and available cardiac-arrest/PCI capability.

## Indications

- Sustained return of spontaneous circulation after adult cardiac arrest.
- Post-arrest patient with a pulse requiring stabilization, monitoring, cause identification, and destination selection.

## Contraindications

- There is no contraindication to immediate post-ROSC stabilization.
- Do not continue AC-10 if the patient becomes pulseless; immediately resume CPR and open the rhythm-appropriate cardiac-arrest protocol.
- Do not give a rapid cold-IV-fluid infusion for prehospital temperature control after ROSC.

## Assessment | Confirm ROSC and prepare for re-arrest

- Confirm pulse, blood pressure/MAP, organized rhythm, perfusion, respiratory effort, and clinical response.
- Keep pads attached and maintain immediate access to the defibrillator, suction, ventilation equipment, and arrest medications.
- Reassess pulse, rhythm, blood pressure, perfusion, $SpO_2$, $EtCO_2$, respiratory status, and neurologic status frequently and after every intervention.

## Assessment | Airway, oxygenation, and ventilation

- Use 100% oxygen until $SpO_2$ can be measured reliably; then titrate $FiO_2$ to maintain $SpO_2$ 90-98%.
- Avoid both hypoxemia and prolonged hyperoxia.
- Begin ventilation near 10 breaths/min and titrate toward $EtCO_2$ approximately 35-45 mm Hg while avoiding hyperventilation.
- Recognize that post-ROSC $EtCO_2$ may underestimate $PaCO_2$ and follow clinical response and blood-gas results when available.
- Confirm an advanced airway with continuous waveform capnography and reassess after every movement.

## Assessment | Hemodynamics

- Obtain frequent blood pressure measurements and maintain MAP at least 65 mm Hg.
- Assess peripheral perfusion, capillary refill, skin findings, mental status, lung sounds, and signs of cardiogenic shock or volume overload.
- Use cautious fluid boluses and transition promptly to norepinephrine when hypotension persists.

## Assessment | Cardiac, neurologic, glucose, and temperature assessment

- Obtain and transmit a 12-lead ECG as soon as feasible; repeat if the tracing was acquired immediately after ROSC or the clinical condition changes.
- Document GCS, pupils, purposeful movement, focal findings, and seizure activity without making early field neurologic prognostication.
- Check blood glucose and treat values below 70 mg/dL through UP-04.
- Measure temperature when available, prevent fever, and avoid rapid warming of spontaneous hypothermia.

## Treatment steps

- Confirm sustained ROSC, keep defibrillation pads attached, and prepare for immediate re-arrest.
- Maintain airway and ventilation, use 100% oxygen until reliable $SpO_2$ is available, and then titrate oxygen to $SpO_2$ 90-98%.
- Begin ventilation near 10 breaths/min and titrate toward $EtCO_2$ approximately 35-45 mm Hg while avoiding hyperventilation.
- Confirm any advanced airway with continuous waveform capnography and reassess breath sounds, chest rise, tube depth, and waveform after every movement.
- Obtain frequent blood pressures and maintain MAP at least 65 mm Hg.
- For MAP below 65 mm Hg, give normal saline 250-500 mL IV/IO and reassess; use smaller volumes for CHF, renal failure, liver failure, or suspected volume overload.
- If hypotension persists, start norepinephrine 0.1-2 mcg/kg/min IV/IO and titrate to MAP at least 65 mm Hg; alternatives require Medical Control.
- Obtain and transmit a 12-lead ECG as soon as feasible and repeat it when the initial tracing was obtained immediately after ROSC or the clinical condition changes.
- Open AC-04 for persistent STEMI/STEMI-equivalent or evidence of significant ongoing myocardial ischemia.
- Check glucose, treat values below 70 mg/dL through UP-04, document neurologic findings and seizure activity, and avoid early neurologic prognostication.
- Measure temperature when available, prevent fever, avoid rapid cold-IV-fluid prehospital cooling, and do not rapidly warm spontaneous hypothermia.
- Transport patients with persistent STEMI/STEMI-equivalent, cardiogenic shock, recurrent ventricular arrhythmias, or significant ongoing ischemia to a PCI-capable hospital and consider Covenant Health Air when it shortens time to definitive care.

- Otherwise transport to the closest appropriate receiving facility with early notification and continuous reassessment.

## Medications

**Normal saline**

250-500 mL IV/IO bolus, then reassess

- AEMT and Paramedic standing order for post-ROSC MAP below 65 mm Hg.
- Use smaller volumes for CHF, renal failure, liver failure, or suspected volume overload.
- Reassess blood pressure/MAP, lung sounds, perfusion, and respiratory status after each bolus.

**Norepinephrine**

0.1-2 mcg/kg/min IV/IO; titrate to MAP at least 65 mm Hg

- Paramedic preferred vasopressor for persistent post-ROSC hypotension after cautious fluid resuscitation.
- Use continuous cardiac monitoring and frequent blood-pressure reassessment.
- Alternative post-ROSC vasopressors require Medical Control.

**Dextrose**

Treat glucose below 70 mg/dL through UP-04 Altered Mental Status

- Avoid hypoglycemia and recheck glucose after treatment.
- Do not treat isolated post-ROSC hyperglycemia routinely in the field unless another protocol indicates treatment.

## Warnings / precautions

- Re-arrest is common; keep pads attached and be prepared to resume CPR and defibrillation immediately.
- Avoid hypoxemia, prolonged hyperoxia, and hyperventilation after ROSC.
- Do not use a rapid cold-IV-fluid infusion for prehospital temperature control.
- Use cautious fluid in CHF, renal failure, liver failure, and suspected volume overload.
- Do not rely on a single immediate post-ROSC ECG to exclude or confirm coronary occlusion; repeat the tracing when indicated.
- Coma alone does not exclude a patient from cardiac-center or catheterization-laboratory activation.
- Do not make early neurologic prognostication in the field.
- Routine prophylactic amiodarone is not required after ROSC without recurrent ventricular dysrhythmia.

## Clinical pearls

- A post-ROSC patient remains critically unstable even when the initial blood pressure and rhythm appear acceptable.
- AHA targets $SpO_2$ 90-98%, normal physiologic ventilation, and MAP at least 65 mm Hg after ROSC.
- Post-ROSC $EtCO_2$ may underestimate $PaCO_2$; use it as a trend and ventilation guide rather than an exact substitute for arterial blood gas.
- Immediate post-ROSC ECG changes may evolve; repeat ECG can reduce both missed occlusion and false activation.
- Persistent STEMI, cardiogenic shock, recurrent ventricular arrhythmia, or ongoing ischemia supports rapid PCI-capable destination.

## Special populations | Unresponsive after ROSC

- Prevent fever and communicate the need for protocolized hospital temperature control between 32°C and 37.5°C.
- Do not initiate prehospital cooling with rapid cold-IV-fluid infusion.

- Avoid early neurologic prognostication and document the examination before sedatives or paralytics when feasible.

## Special populations | Spontaneous hypothermia

- Remove wet clothing and prevent further heat loss.
- Do not rapidly warm a spontaneously hypothermic post-arrest patient.

## Special populations | Post-intubation care

- Open AR-08 for tube/BIAD confirmation, ventilation, sedation, analgesia, and ongoing airway management.
- Reassess tube position and waveform capnography after every movement or transfer.

## Related pathways / contacts

**AC-04 ACS / STEMI** - Open for persistent STEMI/STEMI-equivalent or significant ongoing ischemia.

**AC-06 Narrow-Complex Tachycardia** - Open for recurrent narrow-complex tachycardia after ROSC.

**AC-07 Monomorphic Wide-Complex Tachycardia** - Open for recurrent monomorphic VT with a pulse.

**AC-08 Polymorphic Wide-Complex Tachycardia** - Open for recurrent polymorphic VT with a pulse.

**AC-09 VF / Pulseless VT** - Open immediately if the patient loses pulses in VF or VT.

**UP-04 Altered Mental Status** - Open for glucose below 70 mg/dL or another persistent altered-mental-status cause.

**AR-08 Post-Intubation / BIAD Management** - Open for advanced-airway confirmation, ventilation, sedation, and analgesia.

**AM-03 Dialysis / Renal Failure** - Open for suspected hyperkalemia, dialysis-related arrest, or renal failure.

**TE-07 Overdose / Toxic Ingestion** - Open for suspected toxicologic or overdose-related arrest.

### Source and references

American Heart Association. 2025 Guidelines for CPR and ECC: Post-Cardiac Arrest Care. | American Heart Association. 2025 Adult Post-Cardiac Arrest Care Algorithm. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and related cardiac, airway, medical, renal, and toxicology protocols.

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